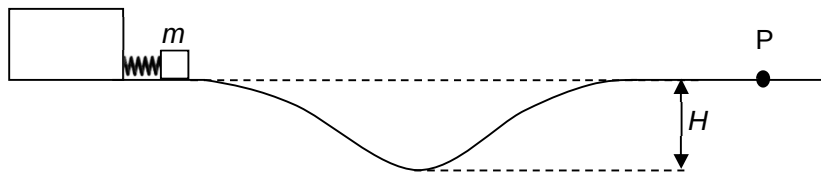


- 7 A small object of mass m is launched by a spring and travels along a rough track to a point P. The spring constant is k and the initial compression of the spring is x . As it travels to P it goes through a dip of depth H . The object experiences a constant force of friction f for the entire distance of d .



What is the kinetic energy of the object at point P?

- A $\frac{1}{2}kx^2 - fd$
- B $\frac{1}{2}kx^2 + fd$
- C $\frac{1}{2}kx^2 + fd + mgH$
- D $\frac{1}{2}kx^2 - fd - mgH$

- 8 A ball rolls down a frictionless circular ramp as shown below.